Your grade: 90%

Your latest: **90%** • Your highest: **90%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. Which of the following measurement equations is/are nonlinear, so requires a nonlinear Kalman filter such as the extended Kalman filter? Select all that apply.

1 / 1 point

- [ ] $z_k = \cos(\pi k/3)x_k + v_k$.
- [x] $z_k = 0.9x_k^3 + 0.3u_k + v_k$.

Correct
Yes. This is not a linear measurement equation because the state's value is cubed.

- [x] $z_k = 4x_k + 2u_k + v_k - 1$.

Correct
Yes, this is nonlinear because of the "-1".

- [x] $z_k = 0.4x_k \times u_k + v_k$.

Correct
Yes. This is not a linear measurement equation because of the multiplication between the state and input.

- [ ] $z_k = 0.9x_k + 0.2u_k$.

2. Consider a system having nonlinear state equation

1 point

$x_k = \frac{1}{4}x_{k-1}^4 + \frac{1}{3}u_{k-1}^3 + \frac{1}{2}w_{k-1}^2$.

If $\hat{x}_{k-1}^+ = 2, u_{k-1} = -1$, and $\bar{w}_{k-1} = 4$, what is $\widehat{B}_{k-1}$? Enter your answer with one digit to the right of the decimal point.

4

Incorrect
No. That is not correct. Confer Lesson 3.1.3.

3. Consider a system having nonlinear measurement equation

1 / 1 point

$z_k = x_k^2 + u_k + v_k$.

If $\hat{x}_k^- = 4, u_k = 4$, and $\bar{v}_k = 0$, what is the value $\hat{z}_k$ computed by an extended Kalman filter? Enter your answer with one digit to the right of the decimal point.

20

Correct
Yes. That is the correct answer.

4. Consider a system having nonlinear measurement equation

1 / 1 point

$z_k = x_k^2 + u_k + v_k$.

If $\hat{x}_k^- = 4, u_k = 4$, and $\bar{v}_k = 0$, what is $\widehat{C}_k$? Enter your answer with one digit to the right of the decimal point.

8

Correct
Yes. That is the correct answer.

5. Consider a system having nonlinear measurement equation

1 / 1 point

$z_k = x_k^2 + u_k + v_k$.

If $\hat{x}_k^- = 4, u_k = 4,\ \bar{v}_k = 0, \Sigma_{\tilde{v}} = 2$ and $\Sigma_{\tilde{x},k}^- = 1$, what is L_k? Enter your answer with two digits to the right of the decimal point.

0.12

Correct
Yes. That is the correct answer.

6. For this problem, consider the nonlinear spring-mass-damper system introduced in Lesson 3.1.5. Assume the following definitions: $m = 40, k_1 = 5, k_2 = 50, d = 0.2, b = 2, \Delta t = 0.2$. Further, assume that $x_k^- = [0,0]^T$ and $x_k^+ = [0,0]^T$.

1 / 1 point

Compute $\hat{B}_k$ for these definitions. What is the (2,1) entry of this matrix? (That is, what is the entry in the second row and first column?) Enter your answer with three digits to the right of the decimal point.

0.005

Correct
Yes, that is correct.

7. In the code presented this week to implement the EKF, which component of the EKF simulator is executed once when the simulation is invoked and initializes simulation constants and variables?

1 / 1 point

- ( ) The wrapper code.
- (•) The function `initEKF.m`.
- ( ) The function `iterEKF.m`.

Correct
Yes, that is correct.

8. For the remaining three questions in this quiz, execute the Coursera Labs Jupyter notebook code for Lesson 3.1.7 and enter results computed by the code.

1 / 1 point

Execute the code as is. What is the rms error for the **velocity** state? Enter your answer with four digits to the right of the decimal point. Note that the code is originally configured to compute the rms error for all the states (position and velocity) so you will need to make minor modifications to compute the desired result.

0.0024

Correct
Yes. That is correct.

9. Once again, for the Jupyter Lab code for Lesson 3.1.7, what is the maximum absolute estimation error for the **position** state? Enter your answer with two digits to the right of the decimal point.

1 / 1 point

0.01

Correct
Yes. That is correct.

10. Once again, for the Jupyter Lab code for Lesson 3.1.7, what is the percentage of time that the **velocity** estimate is outside of its estimation confidence bounds? Enter your answer with one digit to the right of the decimal point.

1 / 1 point

0

Correct
Yes, that is correct.